
BY 501: DNA Nanotechnology

Course Code	BY 501
Course Name	DNA Nanotechnology
L-T-P-C	3-0-0-3
Prerequisites	IC 136 and faculty consent
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	8th Senate (Oct 2015)

Course Contents

- **DNA:**

Brief history about the discovery of the structure of DNA, Concept of DNA helix, Structural features of DNA and its distinction with RNA, Basic concept of gel electrophoresis, DNA amplification and ligation. [6 Lectures]

- **Structural DNA nanotechnology**

Four arm junction, double crossovers, DNA arrays and Lattices, 3D structure- cube, tetrahedron, dodecahedron octahedron, pyramid, DNA origami. [6 Lectures]

- **Dynamic DNA nanotechnology**

Reconfigurable DNA based structures, DNA nanomechanical devices, DNA nanomotors, DNA aptamers, DNA walker, DNA Tweezer, DNzyme – structure function and applications, DNA nanotransport device, molecular cages [10 Lectures]

- **DNA based logic gates**

AND, OR, NOT, XOR, NAND gates, Single and multiple input DNA logic gates, Circular logic gates, DNA Circuits, DNA computing. [10 Lectures]

- **Module-5**

DNA directed Assembly of metal, semiconductor nanoparticles and nanoclusters. DNA Scaffolding, DNA nanorobot, Application of DNA assembled structure in chemical, biological and molecular sensing, DNA-based drug and gene delivery, Future Applications. [10 Lectures]

References

A nascent textbook mentioned below will be used as appropriate and several recent articles from peer reviewed journals like Nature, Science, Nature Nanotechnology, ACS Nano, Nanoletters etc. in the field of DNA nanotechnology will be discussed. The course material will be provided.

1. Edited by Chunhai Fan, **DNA Nanotechnology – From Structure to Function**, Springer-Verlag, 2013.
2. Edited by Giampaolo Zuccheri and Bruno Samorì, **DNA Nanotechnology: Methods and Protocols**, Humana Press, Springer Science+Business Media, 2011.